Drug Consumption (Quantified)

Donated on 10/16/2016

Classify type of drug consumer by personality data

Dataset Characteristics
Multivariate

Subject Area
Social Science

Associated Tasks
Classification

Feature Type
Real

Instances
1885

Features
12

Dataset Information

Additional Information

Database contains records for 1885 respondents. For each respondent 12 attributes are known: Personality measurements which include NEO-FFI-R (neuroticism, extraversion, openness to experience, agreeableness, and conscientiousness), BIS-11 (impulsivity), and ImpSS (sensation seeking), level of education, age, gender, country of residence and ethnicity. All input attributes are originally categorical and are quantified. After quantification values of all input features can be considered as real-valued. In addition, participants were questioned concerning their use of 18 legal and illegal drugs (alcohol, amphetamines, amyl nitrite, benzodiazepine, cannabis, chocolate, cocaine, caffeine, crack, ecstasy, heroin, ketamine, legal highs, LSD, methadone, mushrooms, nicotine and volatile substance abuse and one fictitious drug (Semeron) which was introduced to identify over-claimers. For each drug they have to select one of the answers: never used the drug, used it over a decade ago, or in the last decade, year, month, week, or day.
Database contains 18 classification problems. Each of independent label variables contains seven classes: "Never Used", "Used over a Decade Ago", "Used in Last Decade", "Used in Last Year", "Used in Last Month", "Used in Last Week", and "Used in Last Day".

Problem which can be solved:
* Seven class classifications for each drug separately.
* Problem can be transformed to binary classification by union of part of classes into one new class. For example, "Never Used", "Used over a Decade Ago" form class "Non-user" and all other classes form class "User".
* The best binarization of classes for each attribute.
* Evaluation of risk to be drug consumer for each drug.

Detailed description of database and process of data quantification are presented in E. Fehrman, A. K. Muhammad, E. M. Mirkes, V. Egan and A. N. Gorban, "The Five Factor Model of personality and evaluation of drug consumption risk.," arXiv https://arxiv.org/abs/1506.06297, 2015
Paper above solve binary classification problem for all drugs. For most of drugs sensitivity and specificity are greater than 75%.

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Has Missing Values?
No

Introductory Paper

The Five Factor Model of personality and evaluation of drug consumption risk
By E. Fehrman, A. Muhammad, E. Mirkes, Vincent Egan, A. Gorban. 2015
Published in Data Science

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1 citations
42546 views

Creators

Elaine Fehrman
Vincent Egan
Evgeny Mirkes

DOI

10.24432/C5TC7S

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This allows for the sharing and adaptation of the datasets for any purpose, provided that the appropriate credit is given.

Variables Table

Variable Name	Role	Type	Demographic	Description	Units	Missing Values
id	ID	Integer				no
age	Feature	Continuous	Age			no
gender	Feature	Continuous	Gender			no
education	Feature	Continuous	Education Level			no
country	Feature	Continuous	Nationality			no
ethnicity	Feature	Continuous	Ethnicity			no
nscore	Feature	Continuous				no
escore	Feature	Continuous				no
oscore	Feature	Continuous				no
ascore	Feature	Continuous				no
cscore	Feature	Continuous				no
impuslive	Feature	Continuous				no
ss	Feature	Continuous				no
alcohol	Target	Categorical				no
amphet	Target	Categorical				no
amyl	Target	Categorical				no
benzos	Target	Categorical				no
caff	Target	Categorical				no
cannabis	Target	Categorical				no
choc	Target	Categorical				no
coke	Target	Categorical				no
crack	Target	Categorical				no
ecstasy	Target	Categorical				no
heroin	Target	Categorical				no
ketamine	Target	Categorical				no

Variables Table

Variable Name	Role	Type	Demographic	Description	Units	Missing Values
legalh	Target	Categorical				no
lsd	Target	Categorical				no
meth	Target	Categorical				no
mushrooms	Target	Categorical				no
nicotine	Target	Categorical				no
semer	Target	Categorical				no
vsa	Target	Categorical				no

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Additional Variable Information

1. ID is number of record in original database. Cannot be related to participant. It can be used for reference only.

2. Age (Real) is age of participant and has one of the values:

Value	Meaning	Cases	Fraction
-0.95197	18-24	643	34.11%
-0.07854	25-34	481	25.52%
0.49788	35-44	356	18.89%
1.09449	45-54	294	15.60%
1.82213	55-64	93	4.93%
2.59171	65+	18	0.95%

Descriptive statistics

Min	Max	Mean	Std.dev.
-0.95197	2.59171	0.03461	0.87813

3. Gender (Real) is gender of participant:

Value	Meaning	Cases	Fraction
0.48246	Female	942	49.97%
-0.48246	Male	943	50.03%

Descriptive statistics

Min	Max	Mean	Std.dev.
-0.48246	0.48246	-0.00026	0.48246

4. Education (Real) is level of education of participant and has one of the values:

Value	Meaning	Cases	Fraction
-2.43591	Left school before 16 years	28	1.49%
-1.73790	Left school at 16 years	99	5.25%
-1.43719	Left school at 17 years	30	1.59%
-1.22751	Left school at 18 years	100	5.31%
-0.61113	Some college or university, no certificate or degree	506	26.84%
-0.05921	Professional certificate/ diploma	270	14.32%
0.45468	University degree	480	25.46%
1.16365	Masters degree	283	15.01%
1.98437	Doctorate degree	89	4.72%

Descriptive statistics

Min	Max	Mean	Std.dev.
-2.43591	1.98437	-0.00379	0.95004

5. Country (Real) is country of current residence of participant and has one of the values:

Value	Meaning	Cases	Fraction
-0.09765	Australia	54	2.86%
0.24923	Canada	87	4.62%
-0.46841	New Zealand	5	0.27%
-0.28519	Other	118	6.26%
0.21128	Republic of Ireland	20	1.06%
0.96082	UK	1044	55.38%
-0.57009	USA	557	29.55%

Descriptive statistics

Min	Max	Mean	Std.dev.
-0.57009	0.96082	0.35554	0.70015

6. Ethnicity (Real) is ethnicity of participant and has one of the values:

Value	Meaning	Cases	Fraction
-0.50212	Asian	26	1.38%
-1.10702	Black	33	1.75%
1.90725	Mixed-Black/Asian	3	0.16%
0.12600	Mixed-White/Asian	20	1.06%
-0.22166	Mixed-White/Black	20	1.06%
0.11440	Other	63	3.34%
-0.31685	White	1720	91.25%

Descriptive statistics

Min	Max	Mean	Std.dev.
-1.10702	1.90725	-0.30958	0.16618

7. Nscore (Real) is NEO-FFI-R Neuroticism. Possible values are presented in table below:

Nscore	Cases	Value	Nscore	Cases	Value	Nscore	Cases	Value
12	1	-3.46436	29	60	-0.67825	46	67	1.02119

13	1	-3.15735	30	61	-0.58016	47	27	1.13281
14	7	-2.75696	31	87	-0.46725	48	49	1.23461
15	4	-2.52197	32	78	-0.34799	49	40	1.37297
16	3	-2.42317	33	68	-0.24649	50	24	1.49158
17	4	-2.34360	34	76	-0.14882	51	27	1.60383
18	10	-2.21844	35	69	-0.05188	52	17	1.72012
19	16	-2.05048	36	73	0.04257	53	20	1.83990
20	24	-1.86962	37	67	0.13606	54	15	1.98437
21	31	-1.69163	38	63	0.22393	55	11	2.12700
22	26	-1.55078	39	66	0.31287	56	10	2.28554
23	29	-1.43907	40	80	0.41667	57	6	2.46262
24	35	-1.32828	41	61	0.52135	58	3	2.61139
25	56	-1.19430	42	77	0.62967	59	5	2.82196
26	57	-1.05308	43	49	0.73545	60	2	3.27393
27	65	-0.92104	44	51	0.82562			
28	70	-0.79151	45	37	0.91093			

Descriptive statistics

Min	Max	Mean	Std.dev.
-3.46436	3.27393	0.00004	0.99808

8. Escore (Real) is NEO-FFI-R Extraversion. Possible values are presented in table below:

Escore	Cases	Value	Escore	Cases	Value	Escore	Cases	Value
16	2	-3.27393	31	55	-1.23177	45	91	0.80523
18	1	-3.00537	32	52	-1.09207	46	69	0.96248
19	6	-2.72827	33	77	-0.94779	47	64	1.11406
20	3	-2.53830	34	68	-0.80615	48	62	1.28610
21	3	-2.44904	35	58	-0.69509	49	37	1.45421
22	8	-2.32338	36	89	-0.57545	50	25	1.58487
23	5	-2.21069	37	90	-0.43999	51	34	1.74091
24	9	-2.11437	38	106	-0.30033	52	21	1.93886
25	4	-2.03972	39	107	-0.15487	53	15	2.12700
26	21	-1.92173	40	130	0.00332	54	10	2.32338
27	23	-1.76250	41	116	0.16767	55	9	2.57309
28	23	-1.63340	42	109	0.32197	56	2	2.85950
29	32	-1.50796	43	105	0.47617	58	1	3.00537
30	38	-1.37639	44	103	0.63779	59	2	3.27393

Descriptive statistics

Min	Max	Mean	Std.dev.
-3.27393	3.27393	-0.00016	0.99745

9. Oscore (Real) is NEO-FFI-R Openness to experience. Possible values are presented in table below:

Oscore	Cases	Value	Oscore	Cases	Value	Oscore	Cases	Value
24	2	-3.27393	38	64	-1.11902	50	83	0.58331
26	4	-2.85950	39	60	-0.97631	51	87	0.72330
28	4	-2.63199	40	68	-0.84732	52	87	0.88309
29	11	-2.39883	41	76	-0.71727	53	81	1.06238
30	9	-2.21069	42	87	-0.58331	54	57	1.24033
31	9	-2.09015	43	86	-0.45174	55	63	1.43533
32	13	-1.97495	44	101	-0.31776	56	38	1.65653
33	23	-1.82919	45	103	-0.17779	57	34	1.88511
34	25	-1.68062	46	134	-0.01928	58	19	2.15324
35	26	-1.55521	47	107	0.14143	59	13	2.44904
36	39	-1.42424	48	116	0.29338	60	7	2.90161
37	51	-1.27553	49	98	0.44585			

Descriptive statistics

Min	Max	Mean	Std.dev.
-3.27393	2.90161	-0.00053	0.99623

10. Ascore (Real) is NEO-FFI-R Agreeableness. Possible values are presented in table below:

Ascore	Cases	Value	Ascore	Cases	Value	Ascore	Cases	Value
12	1	-3.46436	34	42	-1.34289	48	104	0.76096
16	1	-3.15735	35	45	-1.21213	49	85	0.94156
18	1	-3.00537	36	62	-1.07533	50	68	1.11406

23	1	-2.90161	37	83	-0.91699	51	58	1.2861
24	2	-2.78793	38	82	-0.76096	52	39	1.45039
25	1	-2.70172	39	102	-0.60633	53	36	1.61108
26	7	-2.53830	40	98	-0.45321	54	36	1.81866
27	7	-2.35413	41	114	-0.30172	55	16	2.03972
28	8	-2.21844	42	101	-0.15487	56	14	2.23427
29	13	-2.07848	43	105	-0.01729	57	8	2.46262
30	18	-1.92595	44	118	0.13136	58	7	2.75696
31	24	-1.77200	45	112	0.28783	59	1	3.15735
32	30	-1.62090	46	100	0.43852	60	1	3.46436
33	34	-1.47955	47	100	0.59042			

Descriptive statistics

Min	Max	Mean	Std.dev.
-3.46436	3.46436	-0.00024	0.99744

11. Cscore (Real) is NEO-FFI-R Conscientiousness. Possible values are presented in table below:

Cscore	Cases	Value	Cscore	Cases	Value	Cscore	Cases	Value
17	1	-3.46436	32	39	-1.25773	46	113	0.58489
19	1	-3.15735	33	49	-1.13788	47	95	0.7583
20	3	-2.90161	34	55	-1.01450	48	95	0.93949
21	2	-2.72827	35	55	-0.89891	49	76	1.13407
22	5	-2.57309	36	69	-0.78155	50	47	1.30612
23	5	-2.42317	37	81	-0.65253	51	43	1.46191
24	6	-2.30408	38	77	-0.52745	52	34	1.63088
25	9	-2.18109	39	87	-0.40581	53	28	1.81175
26	13	-2.04506	40	97	-0.27607	54	27	2.04506
27	13	-1.92173	41	99	-0.14277	55	13	2.33337
28	25	-1.78169	42	105	-0.00665	56	8	2.63199
29	24	-1.64101	43	90	0.12331	57	3	3.00537
30	29	-1.51840	44	111	0.25953	59	1	3.46436
31	41	-1.38502	45	111	0.41594			

Descriptive statistics

Min	Max	Mean	Std.dev.
-3.46436	3.46436	-0.00039	0.99752

12. Impulsive (Real) is impulsiveness measured by BIS-11. Possible values are presented in table below:

Impulsiveness	Cases	Fraction
-2.55524	20	1.06%
-1.37983	276	14.64%
-0.71126	307	16.29%
-0.21712	355	18.83%
0.19268	257	13.63%
0.52975	216	11.46%
0.88113	195	10.34%
1.29221	148	7.85%
1.86203	104	5.52%
2.90161	7	0.37%

Descriptive statistics

Min	Max	Mean	Std.dev.
-2.55524	2.90161	0.00721	0.95446

13. SS (Real) is sensation seeing measured by ImpSS. Possible values are presented in table below:

SS	Cases	Fraction
-2.07848	71	3.77%
-1.54858	87	4.62%
-1.18084	132	7.00%
-0.84637	169	8.97%
-0.52593	211	11.19%
-0.21575	223	11.83%
0.07987	219	11.62%
0.40148	249	13.21%
0.76540	211	11.19%
1.22470	210	11.14%

1.92173 103 5.46%

Descriptive statistics

Min	Max	Mean	Std.dev.
-2.07848	1.92173	-0.00329	0.96370

14. Alcohol is class of alcohol consumption. It is output attribute with following distribution of classes.

15. Amphet is class of amphetamines consumption. It is output attribute with following distribution of classes.

16. Amyl is class of amyl nitrite consumption. It is output attribute with following distribution of classes.

17. Benzos is class of benzodiazepine consumption. It is output attribute with following distribution of classes:

Value	Class	Alcohol		Amphet		Amyl		Benzos	
		Cases	Fraction	Cases	Fraction	Cases	Fraction	Cases	Fraction
CL0	Never Used	34	1.80%	976	51.78%	1305	69.23%	1000	53.05%
CL1	Used over a Decade Ago	34	1.80%	230	12.20%	210	11.14%	116	6.15%
CL2	Used in Last Decade	68	3.61%	243	12.89%	237	12.57%	234	12.41%
CL3	Used in Last Year	198	10.50%	198	10.50%	92	4.88%	236	12.52%
CL4	Used in Last Month	287	15.23%	75	3.98%	24	1.27%	120	6.37%
CL5	Used in Last Week	759	40.27%	61	3.24%	14	0.74%	84	4.46%
CL6	Used in Last Day	505	26.79%	102	5.41%	3	0.16%	95	5.04%

18. Caff is class of caffeine consumption. It is output attribute with following distribution of classes.

19. Cannabis is class of cannabis consumption. It is output attribute with following distribution of classes.

20. Choc is class of chocolate consumption. It is output attribute with following distribution of classes.

21. Coke is class of cocaine consumption. It is output attribute with following distribution of classes:

Value	Class	Caff		Cannabis		Choc		Coke	
		Cases	Fraction	Cases	Fraction	Cases	Fraction	Cases	Fraction
CL0	Never Used	27	1.43%	413	21.91%	32	1.70%	1038	55.07%
CL1	Used over a Decade Ago	10	0.53%	207	10.98%	3	0.16%	160	8.49%
CL2	Used in Last Decade	24	1.27%	266	14.11%	10	0.53%	270	14.32%
CL3	Used in Last Year	60	3.18%	211	11.19%	54	2.86%	258	13.69%
CL4	Used in Last Month	106	5.62%	140	7.43%	296	15.70%	99	5.25%
CL5	Used in Last Week	273	14.48%	185	9.81%	683	36.23%	41	2.18%
CL6	Used in Last Day	1385	73.47%	463	24.56%	807	42.81%	19	1.01%

22. Crack is class of crack consumption. It is output attribute with following distribution of classes.

23. Ecstasy is class of ecstasy consumption. It is output attribute with following distribution of classes.

24. Heroin is class of heroin consumption. It is output attribute with following distribution of classes.

25. Ketamine is class of ketamine consumption. It is output attribute with following distribution of classes:

Value	Class	Crack		Ecstasy		Heroin		Ketamine	
		Cases	Fraction	Cases	Fraction	Cases	Fraction	Cases	Fraction
CL0	Never Used	1627	86.31%	1021	54.16%	1605	85.15%	1490	79.05%
CL1	Used over a Decade Ago	67	3.55%	113	5.99%	68	3.61%	45	2.39%
CL2	Used in Last Decade	112	5.94%	234	12.41%	94	4.99%	142	7.53%
CL3	Used in Last Year	59	3.13%	277	14.69%	65	3.45%	129	6.84%
CL4	Used in Last Month	9	0.48%	156	8.28%	24	1.27%	42	2.23%
CL5	Used in Last Week	9	0.48%	63	3.34%	16	0.85%	33	1.75%
CL6	Used in Last Day	2	0.11%	21	1.11%	13	0.69%	4	0.21%

26. Legalh is class of legal highs consumption. It is output attribute with following distribution of classes

Acid

27. LSD is class of ~~alcohol~~ consumption. It is output attribute with following distribution of classes

28. Meth is class of methadone consumption. It is output attribute with following distribution of classes.

29. Mushrooms is class of magic mushrooms consumption. It is output attribute with following distribution of classes:

Value	Class	Legalh		LSD		Meth		Mushrooms	
		Cases	Fraction	Cases	Fraction	Cases	Fraction	Cases	Fraction
CL0	Never Used	1094	58.04%	1069	56.71%	1429	75.81%	982	52.10%
CL1	Used over a Decade Ago	29	1.54%	259	13.74%	39	2.07%	209	11.09%
CL2	Used in Last Decade	198	10.50%	177	9.39%	97	5.15%	260	13.79%
CL3	Used in Last Year	323	17.14%	214	11.35%	149	7.90%	275	14.59%
CL4	Used in Last Month	110	5.84%	97	5.15%	50	2.65%	115	6.10%
CL5	Used in Last Week	64	3.40%	56	2.97%	48	2.55%	40	2.12%
CL6	Used in Last Day	67	3.55%	13	0.69%	73	3.87%	4	0.21%

30. Nicotine is class of nicotine consumption. It is output attribute with following distribution of classes.

31. Semer is class of fictitious drug Semeron consumption. It is output attribute with following distribution of classes.

32. VSA is class of volatile substance abuse consumption. It is output attribute with following distribution of classes:

Value	Class	Nicotine		Semer		VSA	
		Cases	Fraction	Cases	Fraction	Cases	Fraction
CL0	Never Used	428	22.71%	1877	99.58%	1455	77.19%
CL1	Used over a Decade Ago	193	10.24%	2	0.11%	200	10.61%
CL2	Used in Last Decade	204	10.82%	3	0.16%	135	7.16%
CL3	Used in Last Year	185	9.81%	2	0.11%	61	3.24%
CL4	Used in Last Month	108	5.73%	1	0.05%	13	0.69%
CL5	Used in Last Week	157	8.33%	0	0.00%	14	0.74%
CL6	Used in Last Day	610	32.36%	0	0.00%	7	0.37%

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Dataset Files

File	Size
drug_consumption.data	338.6 KB

Reviews

There are no reviews for this dataset yet.

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